

The Network That Learns It Is a Network: From Predictive Selfhood to Explicit Network Self-Knowledge

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Preprint Version 1.0 - 1 August 2026

Abstract

A nervous system can register its own changing state without possessing an explicit scientific account of what it is. It can also maintain a learned model of its body, history, abilities, confidence, and likely consequences without representing the further proposition that these self-models are produced and revised by a distributed biological network. This paper separates four often-confounded capacities: self-registration, learned self-modeling, embodied first-person perspective, and explicit network self-knowledge. It then asks whether a person can learn a mechanistically accurate account of the network that produces the person's own self-model, transfer that account to unfamiliar cases, and use it to improve calibrated reasoning about perception, emotion, agency, and error. The proposal grows from a dated Self Aware Networks genealogy: a 2011 predictive-self dialogue; public 2012 Neo Mind Cycle discussions of recursive EEG feedback, strange loops, self-prediction, and connectomic organization; a 2017 Neural Lace Podcast account of bidirectional neural communication and receiver-relative pattern difference; 2021 recordings that describe recurrent flow and externally returned brain state; and the mature 2022-2025 SAN architecture. Twenty equations define state transition, self-registration, self-model updating, perspective, mechanistic meta-modeling, counterfactual accuracy, transfer, calibration, mediation, retention, and a conjunctive evidence gate. A preregisterable experiment compares mechanism-first SAN instruction with matched neuroscience facts, philosophical self narratives, and a neutral control. The primary claim is deliberately bounded: reading a theory cannot be assumed to increase consciousness itself. The test is whether instruction produces durable, transferable, perturbation-sensitive, and calibrated explicit knowledge of the distributed mechanism that constructs and revises the self-model. Positive results would establish a measurable form of second-order self-understanding. Null or vocabulary-only results would reject the educational intervention while leaving ordinary self-registration and self-modeling intact. Keywords: self-model; metacognition; predictive processing; network self-knowledge; Self Aware Networks; embodied self; mechanism learning; transfer; calibration; consciousness science

1. The ordinary experience and the scientific problem

You can notice that you are uncertain, recognize your own hand, remember a past decision, feel an emotion building, and revise what you think you saw. These abilities make the self appear immediate. Yet each depends on many partial signals: vision, touch, proprioception, interoception, memory, language, motor prediction, social knowledge, and confidence. No single signal contains the complete person.

The brain therefore faces two different problems. First, it must maintain a useful model of the organism that is sensing and acting. Second, it may sometimes represent an explanation of how that self-model is produced. The first problem is present whenever an organism regulates itself. The second becomes explicit when a person learns that the apparent unity of the self is implemented by distributed, recurrent, state-dependent activity rather than by a hidden inner observer.

Existing research addresses important parts of this picture. Self-model theories explain how an organism can represent itself without positing an additional mental substance [11]. Predictive-processing accounts describe perception and action as inference under a generative model [13,14,29]. Metacognition research measures how well confidence tracks performance [12,15,16,30]. Body-ownership and agency research shows that the experienced boundary and authorship of action can be experimentally altered [17,18]. These accounts do not, by themselves, answer a narrower educational and mechanistic question:

Can a person acquire an accurate model of the distributed network that constructs the person's own self-model, and can that knowledge change how the person reasons about unfamiliar experiences and errors?

The question is not whether a sentence about networks can be memorized. It is whether mechanistic understanding transfers when terminology is removed, whether confidence becomes better calibrated, whether the learner revises the model after contrary evidence, and whether the effect persists.

2. Four capacities that must not be collapsed

2.1 Self-registration

A system self-registers when its own state changes what it does next and when interventions on that state produce discriminable downstream consequences. A thermostat is state-sensitive in a thin sense; an organism can be state-sensitive across many interacting variables. Self-registration alone does not imply a narrative identity, phenomenal consciousness, or an explicit concept of a network.

2.2 Learned self-model

A learned self-model estimates such properties as body location, current goals, likely action outcomes, remembered commitments, competence, fatigue, and social identity. It can be wrong. Rubber-hand and tool-use studies show that bodily representation is dynamically constructed and updated rather than copied from a fixed anatomical map [18,19]. Predictive and interoceptive accounts similarly treat self-representation as an inference constrained by sensory and bodily evidence [13,14,28].

2.3 Embodied first-person perspective

A perspective is the organized relation from which sensory and action variables are evaluated: here rather than there, this body rather than another, this action as mine or not mine. Perspective is not identical to a verbal self-description. Agency research distinguishes a relatively immediate feeling of agency from a later conceptual judgment about agency [17]. The distinction matters because a person can experience ownership or authorship without being able to state the mechanism that generated it.

2.4 Explicit network self-knowledge

Explicit network self-knowledge is the further capacity to represent, test, and revise the proposition that one's self-model and perspective are generated by interacting biological processes. The learner should understand that the model is distributed, recurrent, embodied, temporally extended, and fallible. The learner should also know what the claim does not mean: the person is not unreal; the network is not a separate observer standing behind experience; and naming a mechanism does not dissolve the lived self.

These four levels can dissociate. A person can self-register without explicit theory, have a coherent self-model with poor metacognitive calibration, understand neuroscience vocabulary without transfer, or correctly explain a network mechanism while undergoing an ordinary first-person perspective. The paper therefore refuses a single scalar called "self-awareness."

3. Development of the proposal, 2011-2026

The historical record is used here to show conceptual development, not to claim that every mature SAN term existed at the earliest stage.

Stage	Source-bound contribution	Evidentiary boundary
August 2011	The self is described as a predictive pattern produced across the neocortex; the account rejects a hidden internal witness and treats identity as a context-sensitive organization of learned patterns [1].	The recording date and unlisted-upload date come from owner metadata; the current public watch page displays a 2025 date. This is authorship genealogy, not the strongest adversarial public-priority anchor.

Stage	Source-bound contribution	Evidentiary boundary
January 2012	A Facebook export records Micah Blumberg joining a group named "Self Aware Networks" [5].	The export supports membership and the group name. It does not, by itself, prove who created the group, when the group was created, its public visibility, or a remembered subtitle.
July-August 2012	Public Neo Mind Cycle videos connect Hofstadter's strange loop, EEG feedback, prediction, self-observation, an internally constructed world, connectomic organization, and communication within and between brains [2-4].	These public videos establish the named conceptual ancestry. They do not contain the later NAPOT or PWD terms.
April 2017	The Neural Lace Podcast develops bidirectional read/write communication, spatial neural computation, feedback/no-feedback protocols, and transmission of the difference between a present pattern and an expected pattern [6].	This public source advances the operational architecture but is not yet the complete 2022 SAN formalism.
September-October 2021	Recorder notes describe a recursive brain-state-to-light/sound loop, felt extension into a computer, interruption of returned feedback, and recurrent flow around a center [7,8].	The audio files supply 2021 creation metadata. Statements that an experience occurred in 2011 remain retrospective reports rather than public 2011 fixation.
June-October 2022	Public Git material consolidates Self Aware Networks, NAPOT, receiver-state transformation, neural rendering, tonic/phasic relations, and the network as its own distributed observer [9].	The 2022 Git commits are the earliest strong public fixation for the mature named operator family.
2024-2025	The book and Oscillatory Computational Agency manuscript integrate molecular, oscillatory, embodied, agentic, and artificial-network branches [10,27].	These stages expand and formalize the earlier genealogy; they do not retroactively assign mature wording to 2011 or 2012.
2026	The present paper distinguishes explicit network self-knowledge and specifies controlled educational tests.	The formal measures, experiment, and evidence gate are current contributions and are dated 2026.

The genealogy shows continuity through changing vocabulary. In 2011, the central object is a predictive neural pattern that includes its own changing point of view. In 2012, the public account becomes explicitly recursive: a brain can receive a transformed view of its own activity and learn from the returned signal. In 2017, the architecture becomes a bidirectional communication problem. In 2022, SAN names a multiscale receive-transform-project-feedback system. The present paper asks what changes when that system acquires a scientifically usable description of itself.

4. Established accounts and the remaining gap

4.1 Self-models without a hidden observer

Metzinger's self-model theory supplies an important prior boundary: a system can generate a phenomenal model of itself without requiring an extra entity that possesses the model [11]. Dennett likewise rejects a Cartesian theater in which representations are displayed to a central spectator [22]. SAN agrees with the no-hidden-observer constraint but places additional emphasis on the physical receive-transform-project relations through which neural populations alter one another.

The 2022 SAN recording "Rendering, Not Illusion" explicitly contrasts an illusion-only description with a temporally maintained rendering that the brain both produces and receives [9]. The distinction is not that SAN restores an inner viewer. It is that an implemented rendering has causal work to do: it changes later inference, action, memory, and report. Calling the self a construction does not make it causally inert.

4.2 Predictive selfhood

Predictive-processing and free-energy accounts model perception as inference about the causes of sensory input and action as a way of changing those inputs [13,14,28,29]. The self can enter the model as a set of bodily, perspectival, and controllable causes. The 2011 Blumberg dialogue independently uses prediction as the organizing metaphor for self and world [1], while the 2012 videos connect prediction to recursive feedback and self-observation [2,3]. The present proposal does not claim priority over predictive coding as a field. It identifies the author-specific route by which predictive selfhood became the SAN network architecture.

4.3 Metacognition is measurable but not identical to selfhood

Metacognition research separates first-order task performance from second-order sensitivity: how well confidence distinguishes correct from incorrect judgments [12,15,16]. Prefrontal and interoceptive systems contribute to such judgments, but no single brain region should be treated as a homunculus [12,15]. A person can improve factual knowledge without improving calibration. Conversely, confidence may change while task competence does not. The current experiment must therefore measure content knowledge, transfer, confidence calibration, and model revision separately.

4.4 Philosophical influence is not empirical validation

Hofstadter's strange loops, Dennett's anti-theater arguments, and traditions that question a permanent inner essence helped shape the author's questions [2,21,22]. Hawkins, Sporns, Seung, Buzsaki, and Tse supplied scientific constraints involving prediction, networks, connectomes, oscillations, and choice [23-26]. These works are influences and precursors. They do not validate the current educational effect, and their inclusion does not collapse their theories into SAN.

The remaining empirical gap is therefore narrow: no source cited here establishes that learning a joined, mechanistic account of one's own distributed self-model produces transferable and calibrated second-order self-understanding.

5. The operation before the name

Consider a learner who initially says, "I see the room, I decide, and I feel my emotion." The statement is useful but leaves the mechanism unspecified. A mechanism-first lesson proceeds in six steps.

First, it begins with an ordinary case: the room remains stable while the eyes move, a grip changes when a cup is lighter than expected, or the learner notices anxiety before naming it. Second, it asks what must be accomplished: partial signals have to be combined, assigned to a body and viewpoint, compared with expectations, and used to select an action. Third, it introduces established terms such as multisensory integration, body schema, efference copy, interoception, predictive processing, and metacognition. Fourth, it describes the proposed operation without specialized SAN vocabulary: many receiving populations are changed by incoming signals and by their own prior state; recurrent exchange constrains a temporarily stable estimate; the estimate changes action; returned sensory evidence revises the estimate. Fifth, it names the architecture. Sixth, it identifies measurements and conditions under which the explanation fails.

Self Aware Networks (SAN) is the name used for the wider receive-transform-project-feedback architecture. Neural rendering refers to the causally active, distributed construction that supports a usable scene, body, and perspective. A Phase Wave Differential (PWD) is a proposed receiver-relative difference between an arriving event, the receiver's current state, and an expected relation along a route. Neural Array Projection Oscillation Tomography (NAPOT) names the proposed multiscale reconstruction and re-expression process. None of these terms should appear before the reader understands the problem and the operation they summarize.

The new term introduced by this paper is explicit network self-knowledge: a revisable model, held by the person, of how the person's own distributed neural and bodily processes construct and update the self-model. It differs from merely reciting "the brain is a network." The learner must use the model to predict errors, explain perturbations, identify missing evidence, and revise confidence.

6. Formal model

Let (x_t) denote the organism's distributed internal state, (o_t) incoming sensory and interoceptive evidence, and (a_t) action. The state transition is

$$x_{t+1} = F_{\theta}(x_t, o_t, a_t). \quad (1)$$

A self-registration signal (r_t) is any readout of internal state that can alter later processing:

$$r_t = R_{\rho}(x_t, o_t). \quad (2)$$

Registration is causal only if perturbing a candidate self-variable changes a later response under matched external input:

$$\Delta_{reg} = E[Y | do(r_t = r^+)] - E[Y | do(r_t = r^-)]. \quad (3)$$

The learned self-model (m_t) updates from prior model, registered state, evidence, and action consequence:

$$m_{t+1} = G_{\phi}(m_t, r_t, o_t, a_t, o_{t+1}). \quad (4)$$

For a candidate property (s), the model estimates a posterior conditioned on embodied evidence:

$$p(s_t | o_{1:t}, a_{1:t-1}) = \frac{p(o_t | s_t) p(s_t | o_{1:t-1}, a_{1:t-1})}{Z_t}. \quad (5)$$

Prediction error is a receiver-relative difference between observed and expected evidence:

$$e_t = o_t - E[o_t | m_t, a_{t-1}]. \quad (6)$$

A minimal perspective variable (p_t) combines estimated body state, reference frame, action availability, and sensory ownership:

$$p_t = P(m_t, b_t, f_t, u_t). \quad (7)$$

Action selection depends on the self-model without requiring a central chooser:

$$\pi(a_t | x_t, m_t) = \frac{\exp V(a_t; x_t, m_t)}{\sum_a \exp V(a; x_t, m_t)}. \quad (8)$$

Explicit network self-knowledge adds a meta-model (q_t) of how the self-model is generated:

$$q_t = Q_{\psi}(m_t, K_t), \quad (9)$$

where K_t is acquired mechanistic knowledge. A structural-accuracy score compares the learner's causal graph with a preregistered reference graph:

$$S_{graph} = 1 - \frac{d_{edit}(G_q, G_{\downarrow})}{d_{max}}. \quad (10)$$

Here Z_t is the normalizing constant. Counterfactual accuracy measures whether the learner predicts the consequence of an intervention:

$$S_{cf} = \frac{1}{N} \sum_{i=1}^N I[y_i^{pred}(do(v_i)) = y_i]. \quad (11)$$

Transfer must be measured on examples whose surface vocabulary was absent from instruction:

$$T_{novel} = \mu_{A,novel} - \mu_{A,matched\ control} \quad (12)$$

Metacognitive efficiency can be expressed as second-order sensitivity normalized by first-order sensitivity:

$$M_{eff} = \frac{meta-d'}{d'}. \quad (13)$$

Calibration error for confidence (c_i) and correctness (z_i) is

$$B = \frac{1}{N} \sum_{i=1}^N (c_i - z_i)^2. \quad (14)$$

Define a bounded explicit-network-self-knowledge score from graph accuracy, counterfactual accuracy, novel transfer, and inverse calibration error:

$$K_{ENSK} = w_g S_{graph} + w_c S_{cf} + w_t T_{novel} + w_b (1-B), \quad (15)$$

with weights fixed before data collection. The primary randomized contrast is

$$\tau_K = E[K_{ENSK} | Z=SAN] - E[K_{ENSK} | Z=matched]. \quad (16)$$

To test whether learning mediates transfer rather than demand alone, estimate

$$S_{graph} = f(Z), \quad T_{novel} = g(S_{graph}). \quad (17)$$

Retention after delay (Δt) is

$$R_{\Delta t} = K_{ENSK}(t + \Delta t) - K_{ENSK}(t). \quad (18)$$

A vocabulary-only result is defined by improved term recognition without graph or transfer improvement:

$$V^+ = 1, \quad S_{graph} \approx 0, \quad T_{novel} \approx 0. \quad (19)$$

The strong evidence gate is conjunctive:

$$E_{strong} = AND(A_{acquire}, T_{novel}, C_{calibrate}, R_{retain}, U_{update}). \quad (20)$$

No single questionnaire score can pass this gate.

7. Preregistered experiment

7.1 Participants and allocation

Recruit adults fluent in the instructional language and stratify randomization by prior neuroscience education. Conduct an a priori power analysis using a smallest effect of interest selected before recruitment. Exclude no participant based on whether the participant agrees with SAN. Analyze assigned participants under intention-to-treat and report attrition by condition.

7.2 Four instruction conditions

1. Mechanism-first SAN condition. Ordinary experience, problem, established neuroscience, plain-language receive-transform-project-feedback operation, SAN terms, and discriminating tests.
2. Matched neuroscience condition. The same duration, reading level, examples, diagrams, and established facts, without the joined SAN architecture or explicit network-self thesis.
3. Philosophical self condition. Matched material on no-self, strange loops, and narrative identity, without the neural mechanism.
4. Neutral active control. Matched instruction on an unrelated scientific system with comparable causal complexity.

Instructor enthusiasm, visual quality, time on task, testing frequency, and mention of uncertainty must be matched. Materials should avoid telling participants that one condition is expected to improve self-understanding.

7.3 Measurement sequence

Before instruction, collect first-order performance, confidence, graph construction, counterfactual reasoning, interoceptive and agency judgments, and prior beliefs. Immediately after instruction, repeat parallel forms. At delayed follow-up, use new cases and an error-correction task. The decisive tests are not definitions but transfer problems such as:

- why a stable room can be experienced during eye movements;
- how a mistaken body-ownership judgment can arise without a hidden viewer;
- why high confidence can coexist with low accuracy;
- how emotional bodily evidence can influence a decision without equating emotion with voltage;
- how a distributed system can revise an action without a central executive self;
- what observation would contradict the proposed receive-transform-project-feedback account.

7.4 Perturbation-sensitive updating

Participants receive a case that conflicts with their initial causal graph. One version shows a self-report changing while first-order performance stays fixed; another shows performance changing without an ownership judgment. The learner must identify which edge of the model should be revised. Correct update after contrary evidence is more diagnostic than agreement with the original lesson.

7.5 Primary and secondary endpoints

The primary endpoint is K_ENSK at delayed transfer, with weights and scoring rubric preregistered. Secondary endpoints are graph accuracy, counterfactual accuracy, meta-d-prime efficiency, Brier calibration, retention, mechanistic explanation quality, and update accuracy. Exploratory endpoints may include changes in interoceptive interpretation, agency judgment, or emotional labeling. No exploratory outcome should be redescribed as increased consciousness.

8. Analysis and identifiability

8.1 Separate knowledge from allegiance

Participants can correctly explain a theory they reject. Scoring should reward causal accuracy, boundary recognition, and falsifier identification, not endorsement. A blind rubric should score de-identified responses. Agreement with the sentence "I am a network" is not an endpoint.

8.2 Separate terminology from mechanism

Half of transfer items should omit SAN vocabulary. A participant who learned only the acronyms should fail those items. A second set should use deliberately unfamiliar terms for the same operation. Success across paraphrase tests functional understanding rather than phrase matching.

8.3 Control for general scientific instruction

The matched-neuroscience condition is essential. If both scientific conditions improve equally, the effect belongs to general neuroscience education, not the joined SAN explanation. If the philosophy condition changes attitudes but not mechanistic transfer, the measures have separated narrative impact from causal understanding.

8.4 Mediation and robustness

Estimate whether graph learning mediates novel transfer. Repeat analyses with preregistered and equal weights, robust estimators, multiple-imputation sensitivity, and a per-protocol supplement. Report all conditions and endpoints. Correct for the secondary-outcome family and publish the scoring rubric before unblinding.

8.5 Manipulation and demand checks

Measure perceived novelty, plausibility, emotional impact, instructor confidence, and guessed hypothesis. A result that disappears after accounting for demand is not strong evidence. Nor is a result driven only by participants already committed to a physicalist or contemplative worldview.

9. Predictions and failure conditions

The paper makes five ranked predictions.

1. Mechanism-first instruction should improve causal-graph accuracy relative to all controls.
2. The improvement should transfer to unfamiliar examples whose vocabulary differs from instruction.
3. Confidence should become better calibrated, not merely higher.
4. Learners should revise the correct part of the model after contrary evidence.
5. A portion of the effect should remain at delayed follow-up.

The proposal fails in informative ways:

- Vocabulary failure: terms are remembered but novel transfer remains at control level.
- Generic-education failure: SAN and matched neuroscience produce indistinguishable improvement.
- Persuasion failure: endorsement rises without graph accuracy or calibration.
- Rigidity failure: learners repeat the architecture but cannot revise it after a counterexample.
- Retention failure: immediate scores rise but delayed scores collapse.
- Construct failure: independent raters cannot reliably distinguish graph accuracy, transfer, and endorsement.

A null result does not show that brains lack self-models. It rejects the tested lesson, duration, population, or claim that this form of explicit mechanistic instruction produces the specified second-order change.

10. What is distinctive, and what is inherited

The paper does not rename predictive coding, metacognition, body schema, or self-model theory. Those fields supply established constructs and measurements. The SAN-specific contribution is the joined route from receiver-state transformation and recurrent neural rendering to an explicit, teachable model of the network that constructs the learner's self-model.

The distinction matters because older terms answer different questions. Predictive coding specifies a family of inference and error-minimization schemes. Metacognition measures knowledge about performance or mental states. Body-schema research tests dynamic bodily representation. Self-model theory explains subjectivity without a substantial inner self. SAN's present proposal asks whether these components can be learned as one operational architecture and then used by the same network to reason about its own construction.

This is second-order in a precise sense: the self-model represents the organism; the meta-model represents how that self-model is generated and revised. It is not higher because it is spiritually superior, and it is not proof of a new level of phenomenal consciousness.

11. Limits, ethics, and interpretation

The historical sources include public videos, public Git commits, owner metadata, a private Facebook export, and Recorder files. These have different evidentiary grades. The paper preserves those boundaries rather than assigning one date to the entire theory. Private source material is not made public by citation; public release requires an author-controlled source packet with only the evidence necessary to support each statement.

The intervention should not be marketed as therapy, awakening, enlightenment, or a cure for distress. Learning that the self is constructed can be clarifying for some people and destabilizing for others. Participants must be told that a constructed self can be biologically real, causally efficacious, socially responsible, and worthy of care. Adverse reactions, depersonalization-like discomfort, and withdrawal should be monitored.

No result licenses a claim that SAN uniquely explains all selfhood or that philosophical and contemplative traditions are isomorphic to a neural mechanism. Formal correspondences may be proposed at the level of relations, but equivalence requires explicit objects, mappings, preserved operations, and failure cases.

Finally, educational change is not ontological proof. Even a successful experiment would show that a particular mechanistic account can be learned and used. It would not, by itself, prove every molecular, oscillatory, NAPOT, PWD, or consciousness claim in the larger SAN program.

12. Conclusion

A network can be altered by its own state, maintain a model of its body and history, and organize a first-person perspective without explicitly knowing that it is a network. The next question is whether the network can learn a correct, usable, and revisable account of that fact.

This paper converts that question into a controlled research program. It distinguishes four capacities, preserves the 2011-2025 genealogy without backdating mature terminology, defines mechanistic and metacognitive measures, and requires transfer, calibration, updating, and retention. The proposal succeeds only if the learner gains more than a slogan. The network must demonstrate that it has learned what it is by using the model when the words, examples, and expectations change.

Author contributions and AI-assisted development

Micah Blumberg conceived the Self Aware Networks framework, the historical genealogy, the four-level distinction, and the explicit-network-self-knowledge hypothesis. AI tools assisted with source routing, bibliographic checking, formalization, adversarial review, and editorial production under the author's direction. All scientific claims, interpretations, and publication decisions remain the author's responsibility.

Data and materials availability

This is a theory-and-methods paper. No participant data were collected. The final preprint package contains a claim-source-genealogy atlas, source manifest, figure plan, bibliography, and validation receipt. Public release of private-source excerpts is not authorized by this package.

Ethics statement

No human-subject experiment was conducted for this paper. Any future experiment must receive appropriate institutional ethics review, informed consent, adverse-event monitoring, privacy protection, and preregistration before recruitment.

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